HYPERERY REI (HYPERSONICS REIMAGINED)

Research Dissertation Summary: Computational Aerothermodynamics & Optical Phenomenolgy

ABSTRACT

This research investigates two critical domains in high-speed aerospace engineering: **high-altitude rocket plume illumination geometry** via atmospheric refraction modeling, and **second-order spatial adaptation** using Hessian matrix eigenvalue analysis for high-gradient flow resolution. Utilizing numerical simulation frameworks in Python and Google Colab, this study establishes precise visibility criteria for launch tracking and lays the mathematical groundwork for robust aerodynamic mesh optimization.

SECTION 1: Atmospheric Solar Geometry & Plume Illumination (CRS-35 Case Study)

- **Observer-Site Elevation Tracking:** Evaluated time-dependent solar elevation angles across multiple geographic observation stations in Georgia (Cleveland, Gainesville, and Locust Grove) relative to mission liftoff schedules. Results demonstrate that ground-based observers operate under negative solar elevation limits $(\approx -2^\circ \text{ to } -14^\circ)$ during twilight launch windows.

- **Solar Depression vs. Minimum Altitude Thresholds:** Mapped the geometric boundary conditions required for direct solar illumination of high-altitude vehicle exhaust plumes. Quantitative analysis confirms that observing plumes at altitudes ranging from 50 km to 200 km requires overcoming specific solar depression angle limits $(7^\circ \text{ to over } 14^\circ)$.

- **Atmospheric Refraction & Transmission:** Integrated the Beer-Lambert law and ray-tracing equations $(\frac{d}{ds}\left(n\frac{dr}{ds}\right) = \nabla n)$ to quantify infrared radiance attenuation and apparent versus geometric elevation shifts caused by density gradients in lower atmospheric layers.

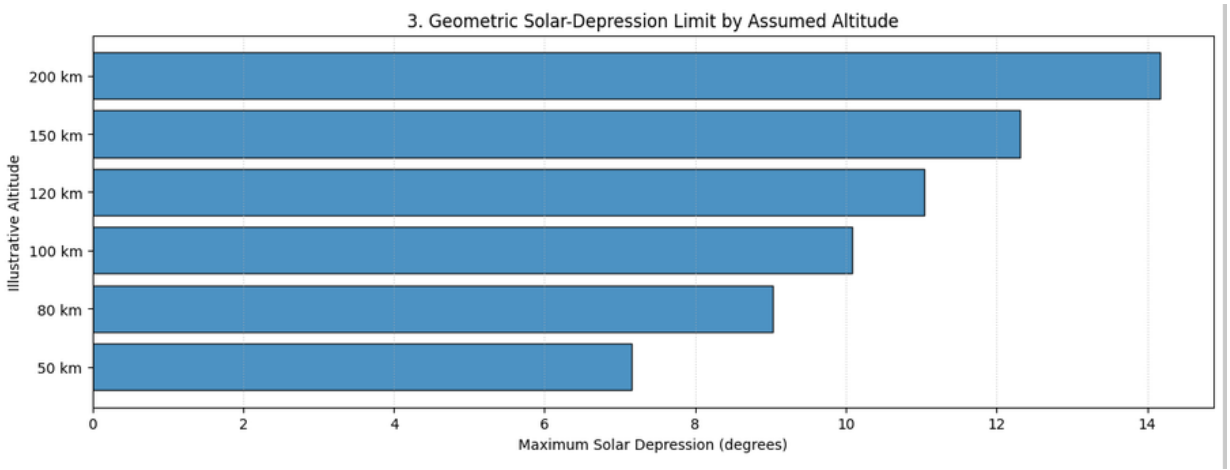
SECTION 2: Hessian-Based Quadratic Forms & Anisotropic Mesh Adaptation

- **Second-Order Curvature Analysis:** Implemented Hessian matrix formulations ($H = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$) to analyze spatial error distribution and directional flow gradients in computational fluid dynamics (CFD) domains.

- **Saddle-Surface Eigenvalue Decomposition:** Utilized directional eigenvectors corresponding to positive-curvature and negative-curvature domains to construct analytical saddle surfaces. This mathematical approach isolates severe shock fronts, boundary layer separations, and density fluctuations that induce optical wavefront distortion.

- **Algorithmic Implication:** Translating these Hessian eigen-structures into automated grid adaptation pipelines allows for precise, anisotropic mesh refinement, minimizing computational grid overhead while maximizing shock-capture fidelity.

Google Colab



Geometric Solar-Depression Limit by Assumed Altitude

Core Concept

This chart calculates the maximum solar depression angle (how far the sun is below the horizon) at which an object—such as a rocket plume or high-altitude vehicle—can still be hit by direct sunlight at various altitudes.

Key Technical Insights

- **Horizon Shadowing:** Even when a ground observer is in total darkness (the sun is below the horizon), objects high in the atmosphere can still catch direct sunlight because they sit above the Earth's shadow line.
- **Altitude-Dependent Visibility Cutoffs:**
    - **At 50 km Altitude:** Direct sunlight reaches the plume only if the solar depression angle is within approximately 7.2 degrees.
    - **At 100 km Altitude (Kármán Line):** The maximum solar depression threshold expands to about 10.1 degrees.
    - **At 200 km Altitude:** Direct sunlight can illuminate the plume even when the sun is as far as 14.2 degrees below the horizon.

Engineering Significance

This geometric boundary is critical for determining whether an exhaust plume is **sunlit (scattered solar radiation)** versus **thermally self-emitting (infrared/chemiluminescence)**. Knowing these exact angular limits allows tracking algorithms to properly predict optical visibility, sensor band selection, and background contrast during twilight launch trajectories.

Atmospheric Optics & Trajectory Illumination Analysis

1. Chart Breakdown: Idealized Minimum Altitude for Direct Sunlight

- **Core Concept:** This continuous curve models the exact mathematical relationship between the **solar depression angle** at a given point of interest and the **minimum altitude** required for a vehicle or plume to clear the Earth's horizon and intercept direct sunlight.
- **Technical Insights:**
    - The curve demonstrates a non-linear (quadratic) growth pattern: as the sun dips further below the horizon (increasing solar depression from 0 to 15 degrees), the altitude required to catch direct solar rays scales from ground level (0 km) up to over 225 km.
- **Engineering Significance:** Provides a rapid lookup or analytical constraint for mission planners to determine whether a rocket body or exhaust plume is operating in direct sunlight or shadowed twilight during ascent stages.

2. Chart Breakdown: Solar Elevation at Observer Sites vs. Time After Liftoff (CRS-35 Case Study)

- **Core Concept:** Tracks the time-dependent solar elevation angle across multiple ground-based observation stations in Georgia (**Cleveland**, **Gainesville**, and **Locust Grove**) over a 60-minute post-liftoff window.
- **Technical Insights:**
    - All three observation sites experience nearly identical solar elevation trajectories due to their geographic proximity, starting at approximately –14.2 degrees at liftoff (t = 0) and steadily climbing linearly to roughly –2 degrees at t = 60 minutes.
    - Because all values remain strictly below the 0 degrees **geometric horizon** dashed line, the ground observers are positioned in twilight/darkness while tracking high-altitude flight events.
- **Engineering Significance:** Establishes the baseline observation geometry for twilight launches, confirming that ground-based optical sensors are capturing scattered plume radiance against a dark background rather than daytime ambient sky glare.

Hessian Matrix Eigenstructure & Anisotropic Mesh Adaptatio

SECTION: Hessian Matrix Eigenstructure & Anisotropic Mesh Adaptation

1. Technical Breakdown: Hessian Quadratic Form — Saddle Surface Analysis

- **Core Concept:** Displays a 3D saddle surface generated by a Hessian matrix possessing mixed-sign eigenvalues $(\lambda_1 > 0, \lambda_2 < 0)$.
- **Mathematical Foundation:** The local behavior of a scalar field $f(x, y)$ (e.g., fluid density or optical refractive index) near a critical point is governed by the quadratic form:

$$f(u, v) \approx \frac{1}{2}\left(\lambda_1 u^2 + \lambda_2 v^2\right)$$

where u and v are coordinates aligned along the principal eigenvectors of the Hessian matrix H.

- **Technical Insights:**

    - **Positive-Curvature Eigenvector (u-axis):** Demonstrates upward local curvature (lambda_1 > 0).
    - **Negative-Curvature Eigenvector (v-axis):** Demonstrates downward local curvature (lambda_2 < 0), creating the characteristic hyperbolic saddle profile.
- **Engineering Significance:** In hypersonic plume modeling, saddle surfaces represent complex shock interactions, shear layer boundaries, and expansion fans where spatial gradients vary drastically depending on direction. Identifying these principal directions allows numerical solvers to align mesh elements anisotropically along the local flow topology.

2. Technical Breakdown: Quadratic Surface Defined by Hessian Matrix H = $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

- **Core Concept:** Maps the 3D surface generated by the specific 2 x 2 symmetric Hessian tensor H = $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$ over the spatial domain $(x, y) \in [-3, 3] \times [-3, 3]$.
- **Mathematical Foundation:**

The surface equation evaluated by the code is given by:

$$f(x,y) = \frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x^2 + 3xy + 2y^2$$

- **Technical Insights:**
    - **Determinant & Eigenvalue Check:** det(H) = (2)(4) - (3)(3) = -1 < 0. Because the determinant is negative, the Hessian matrix is indefinite, producing an elongated saddle surface with a critical inflection point at (0,0,0).
    - The strong cross-term (3xy) rotates the principal axes relative to the standard Cartesian (x,y) grid.
- **Engineering Significance:** This formulation forms the core of **Hessian-based anisotropic mesh adaptation algorithms**. By computing the Hessian tensor of flow variables (such as Mach number or gas density) at each CFD cell, the solver automatically stretches grid elements along directions of low curvature while refining grid resolution along directions of extreme high curvature, drastically reducing computational cost while capturing crisp shock discontinuities.

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